

INTRODUCTION TO DATA MANAGEMENT PROJECT REPORT

(Project Semester January-May 2025)



INDIAN PREMIER LEAGUE

Submitted by

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DECLARATION

I, Lokesh Kumar student of BTECH at, Lovely Professional University, Punjab, hereby declare that all the information furnished in this project report is based on my own intensive work and is genuine.

Date:12-04-2025

Registration No :12302828

ACKNOWLEDGEMENT

I would like to express my special thanks of gratitude to my teacher Dr.Mrinalini Rana who gave me the golden opportunity to do this wonderful project of analysis of the data of a superstore namely "IPL DATA ANALYSIS" which also helped me in doing a lot of research and I came to know about so many new things. I am thankful to them. Secondly, I would also like to thank my parents and friends who helped me a lot in finalizing this project within the limited time frame.

TABLE OF CONTENT

1. Introduction
2. Scope of the Analysis
3. Source of dataset
4. ETL process
5. Analysis on dataset (for each analysis)
 1. Introduction
 2. General Description
 3. Specific Requirements, functions and formulas
 4. Analysis results
 5. Visualization
6. List of Analysis with results
7. References
8. Bibliography

INTRODUCTION

Data Analysis is a process of inspecting, cleansing, transforming, and modeling data with the goal of discovering useful information, informing conclusions, and supporting decision-making. Data analysis has multiple facets and approaches, encompassing diverse techniques under a variety of names, while being used in different business, science, and social science domains.

The analytics team of INDIAN PREMIER LEAGUE (IPL) association and BCCI association anywhere in the world would love to check out through data analysis of each and every match leading to a well-organized and fruitful information. My analysis contains data on host teams, all stadiums, most supported team and teams which are most successful.

The Indian Premier League (IPL), is a Professional 20-20 cricket league in India contested during April and May of every year by teams representing Indian cities and some states. The league was founded by the Board of control of cricket India (BCCI) in 2008, and is regarded as the brain child of Lalit Modi, the founder and former commissioner of the league. IPL has an exclusive window in ICC Future Tours Programs. The IPL is the most-attended cricket league in the world and in 2014 ranked sixth by average attendance among all sports leagues. In 2010, the IPL became the first sporting event in the world to be broadcast live on youtube. The brand value of IPL in 2017 was US\$6.3 billion. according to Duff and help. According to BCCI, the 2015 IPL season contributed 11.5 billion (US\$182 million) to the GDP of the Indian economy.

There have been eleven seasons of the IPL tournament. The current IPL title holders are the CSK, who won the 2018 season. The most successful franchises in the tournament are the CSK and MI with 3 tournament wins each.

Prize Money:

The 2015 season of the IPL offered a total prize money of 240 crore (US\$5.6 million), with the winning team netting 15 crore (US\$2.1 million). The first and second runners up received 10 and 7.5 crores, respectively, with the fourth placed team also winning 7.5 crores.

The others teams are not awarded any prize money. The IPL rules mandate that half of the prize money must be distributed among the players.

Rules:

IPL games utilise television timeouts and hence there is no time limit in which teams must complete their innings. However, a penalty may be imposed if the umpires find teams misusing this privilege. Each team is given a two-and-a-half-minute "strategic timeout" during each innings; one must be taken by the bowling team between the ends of the 6th and 9th overs and one by the batting team between the ends of the 13th and 16th overs.

Insights of data:

INDIAN PREMIER LEAGE ANALYSIS contains the following data fields: -

- Season - The year on which the following match is held.
- Date Contains date on which the match is held.
- Venue:Stadium in which match is held.
- City in which match is held.
- Team 1,Team2
- Toss-Winner Who is the toss winner.
- Toss Decision What team decided after winning the toss.
- Winner - Who won the match.
- Player of the match-Who won the player of the match.
- Umpire-1-Name of Umpire 1.
- Umpire-2-Name of Umpire 2.
- ID This column show the match number.

SCOPE OF ANALYSIS

This project on Indian Premiere League Statistics of India provides the overall Statistics details of the matches of IPL and teams progress in various aspects from the year 2008 to 2018.

Objectives of this project:

- To good hand on excel.
- To use different feature and get friendly with the excel.
- How to link one sheet to another and traverse between different sheets.
- How to use pivot table and pivot chart.
- Learn to make dashboard in excel.
- To make different type of graphs in excel.
- To learn how to fetch data from other source to excel in different formats.

Aim of this project is to answer the above objectives in the form of visualization by creating a dashboard to convey the answers effectively and efficiently..

SOURCE OF DATASET

The data is being taken from the GITHUB It is an AirBnB for Data Scientists - this is where they spend their nights and weekends. It's a crowd-sourced platform to attract, nurture, train and challenge data scientists from all around the world to solve data science, machine learning and predictive analytics problems. It has over 536,000 active members from 194 countries and it receives close to 150,000 submissions per month. Started from Melbourne.

EDA PROCESS

In computing, extract, transform, load (ETL) is a process in database usage to prepare data for analysis, especially in data warehousing. Data extraction involves extracting data from homogeneous or heterogeneous sources, while data transformation processes data by transforming them into a proper storage format structure for the purposes of querying and analysis; finally, data loading describes the insertion of data into the final target database such as an operational data store, a data mart, or a data warehouse. A properly designed ETL system extracts data from the source systems, enforces data quality and consistency standards, conforms data so that separate sources can be used together, and finally delivers data in a presentation-ready format so that application developers can build applications and end users can make decisions.

Precisely, ETL is defined as a process that extracts the data from different RDBMS source systems, then transforms the data (like applying calculations, concatenations, etc.) and finally loads the data into the Data Warehouse system. ETL stands for Extract, transform, load.

Cleaning:

Removing the blank cells from the dataset

Removing Columns which are not properly defined or not Crucial to our analysis

Giving Proper and appropriate Column names

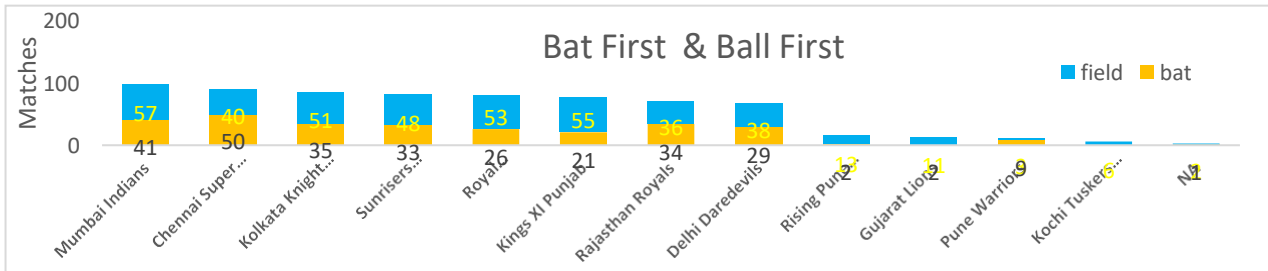
Excluding the Null values from the data

Improving data formatting

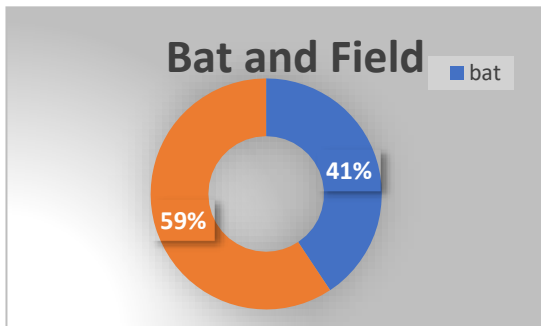
Removing duplicate values

Analysis of Dataset

Team Decision :It is Based on Bat and Field First Wins For each and every Team .

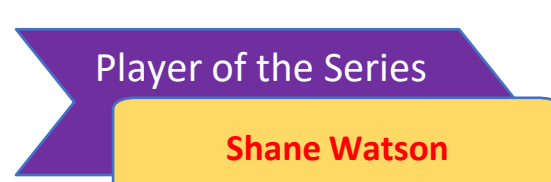


Toss Decision:Team win based on bat and field

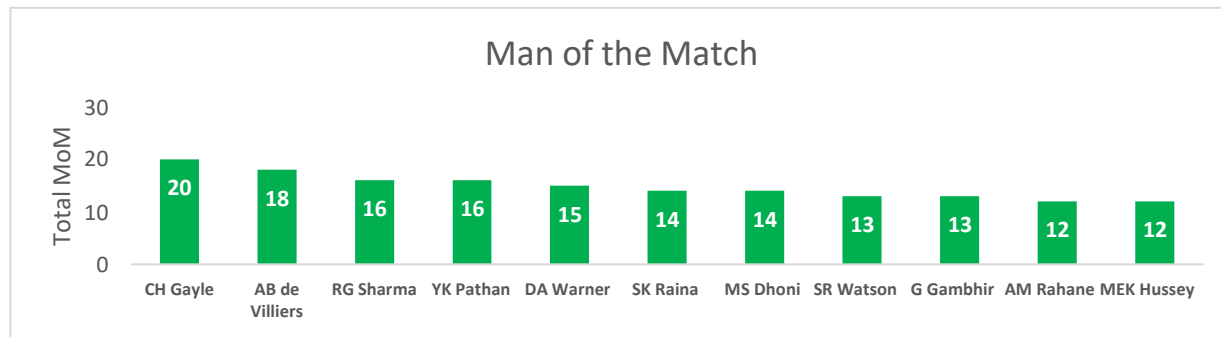


Header :It consist of Season,Winner,Runner UP,Player of Series.It changes Dynamically According to season.

Formulas used:VLOOKUP

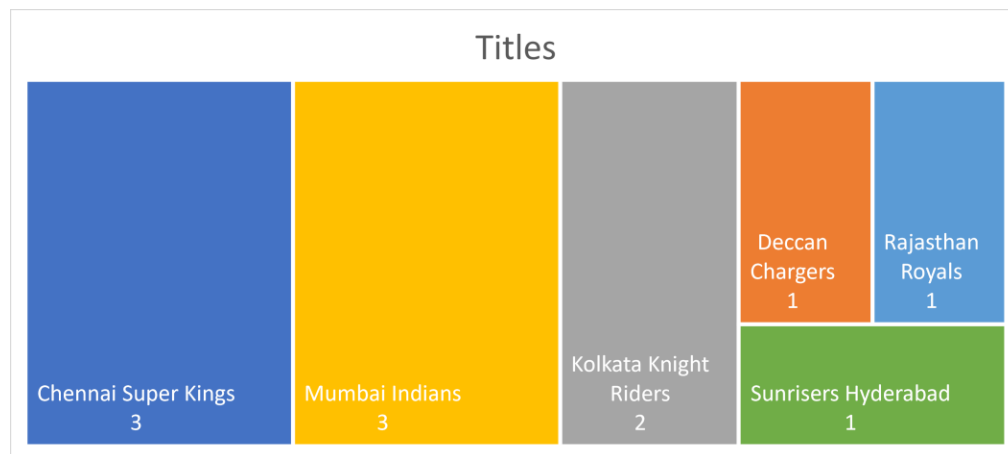


Man of the Matches: It given details that top 10 man of match award winners from 2008 to 2018

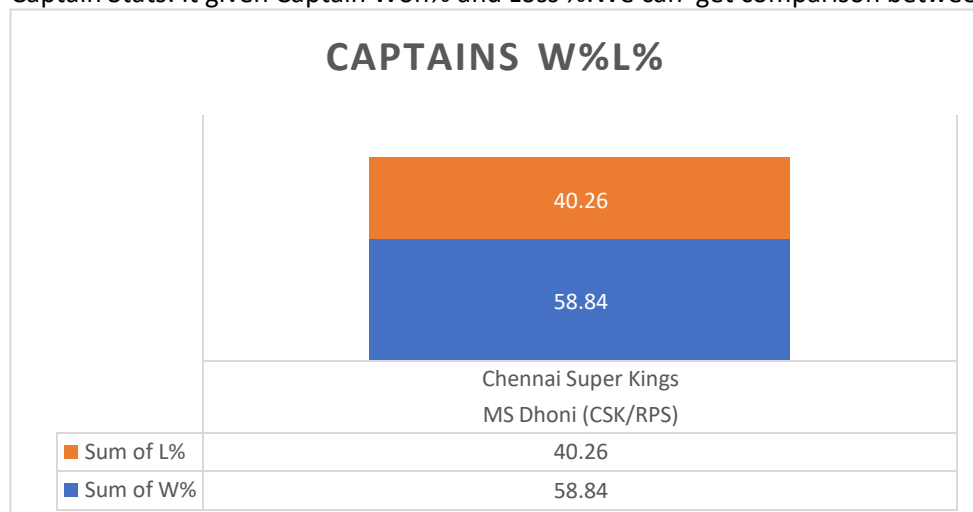


Titles :It gives that till 2018 How many teams won titles and their count.

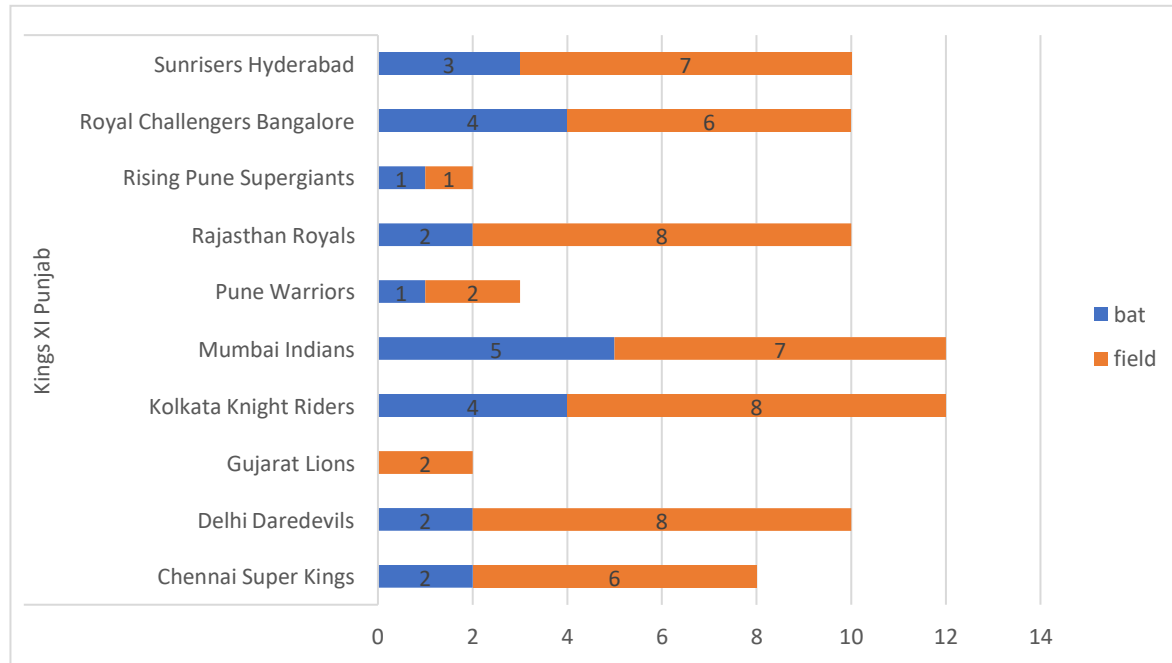
Formula used: =GETPIVOTDATA()



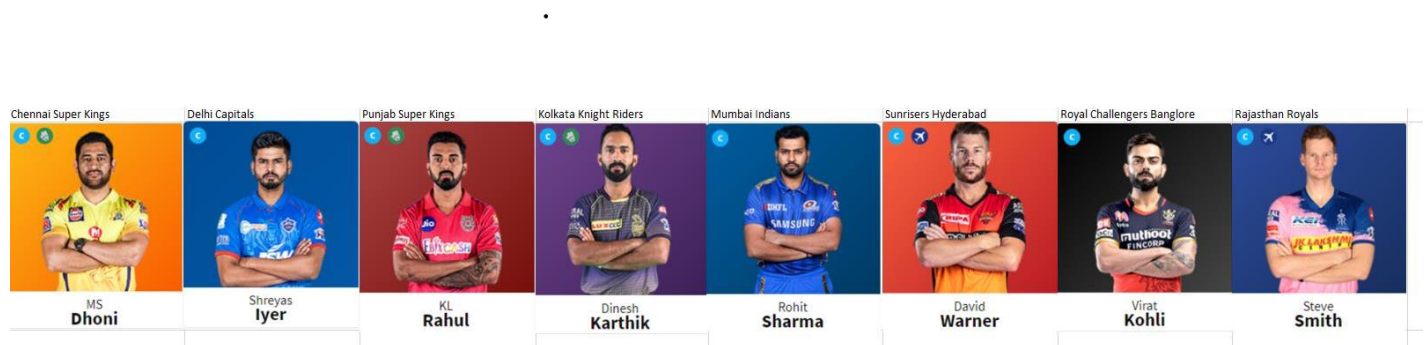
Captain Stats: It given Captain Won% and Loss %.We can get comparison between two or more captains

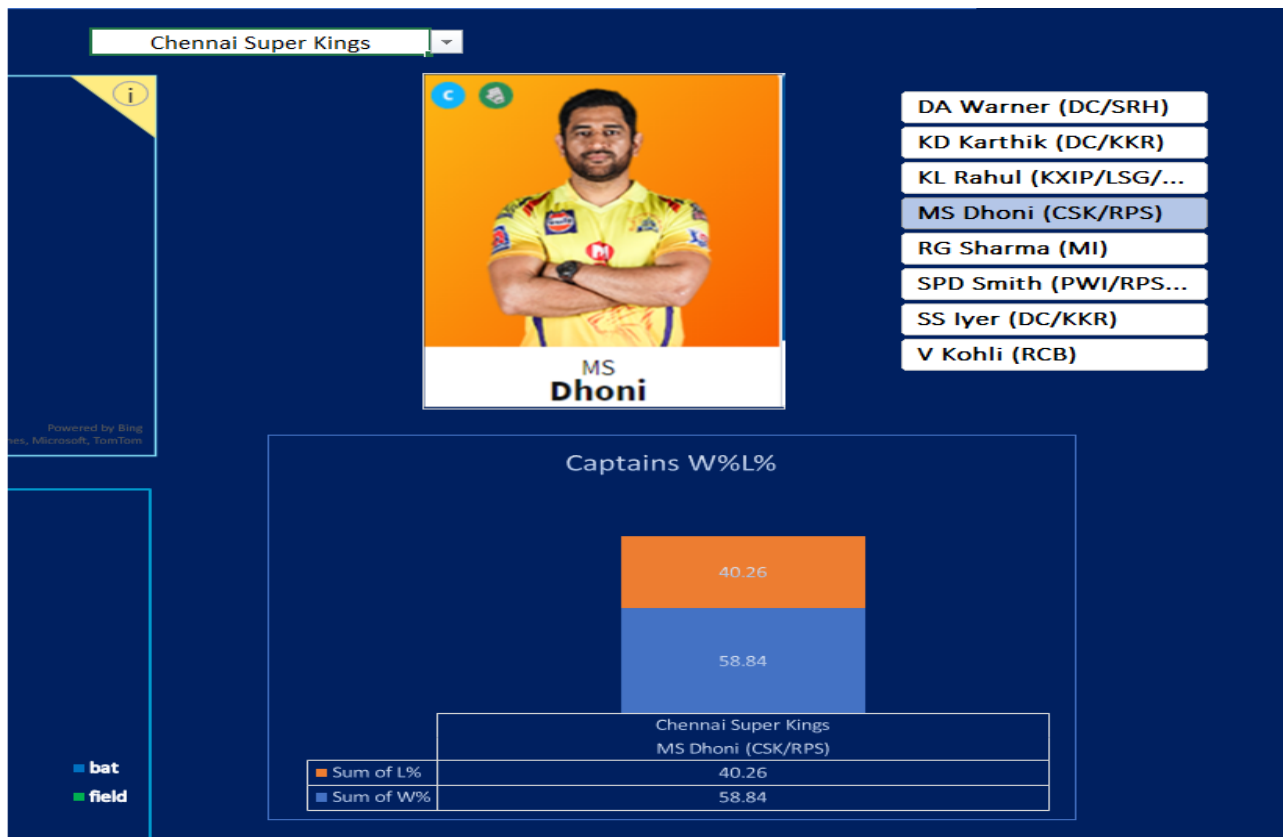


Count of Winner: Team 1 vs Team 2 who won and their toss choices.



Captains search :It gives captain Image when we select team from dropdown list. Captains are 2018 year.





Formulas Used: INDEX(),MATCH(),LIST VALIDATION

KEY FEATURES:

Hyperlinks to connect two sheets of dashboard.

Macros To move heading of IPL.

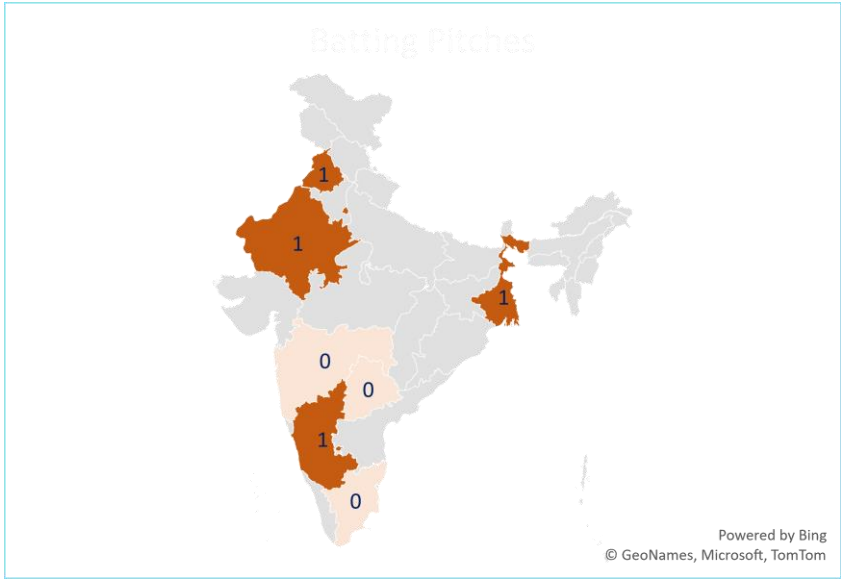
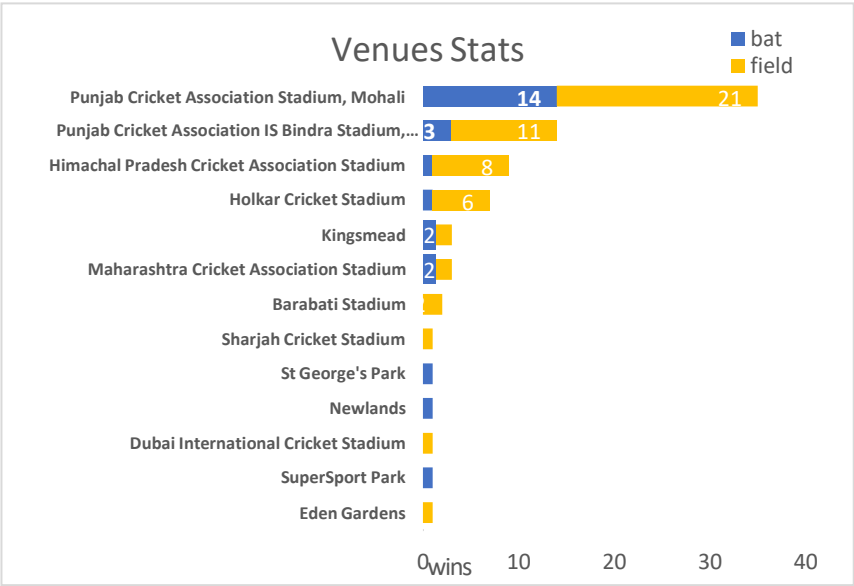
Fixing Image to particular Cell.

Drop down list to select team captains.

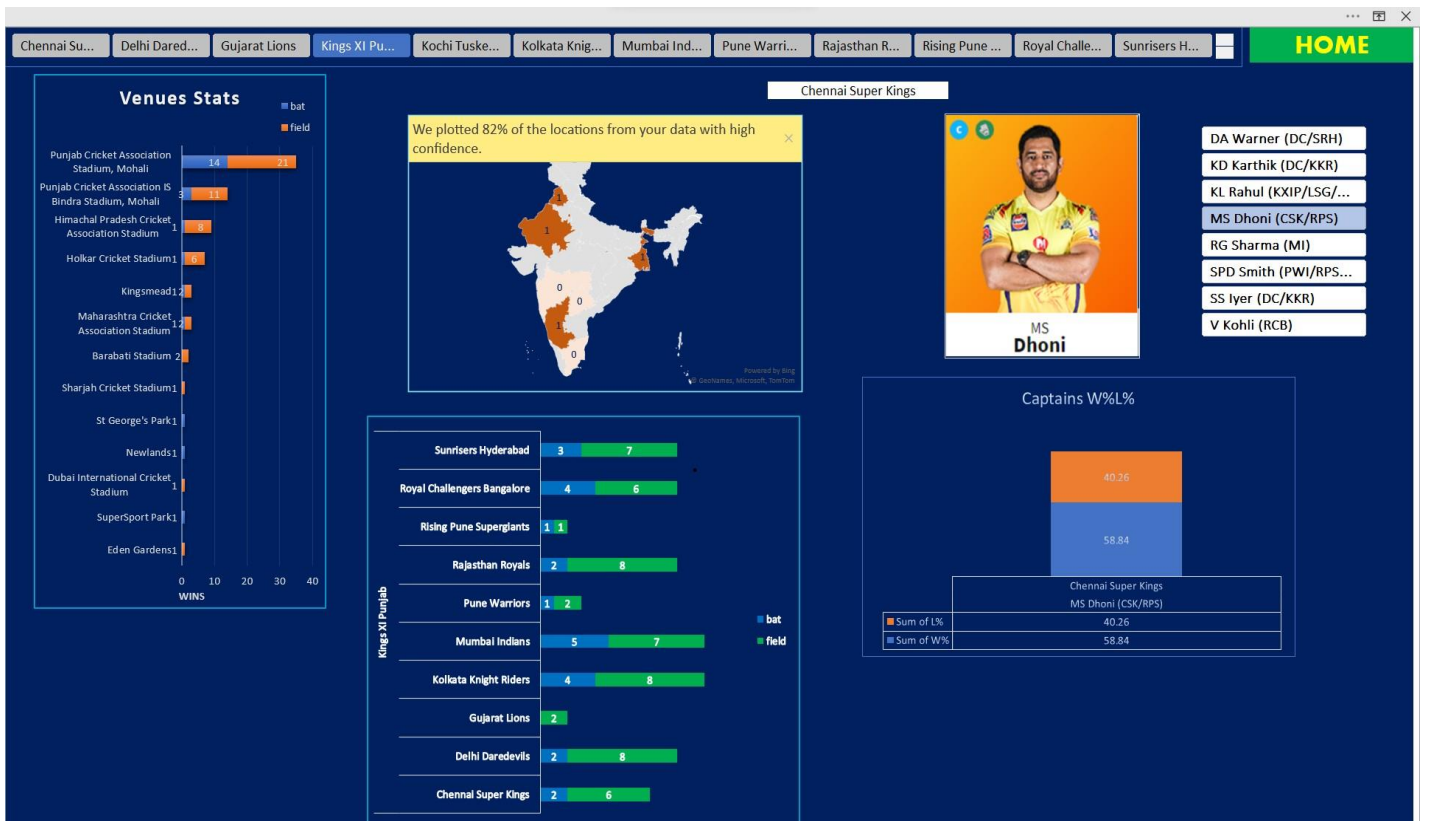
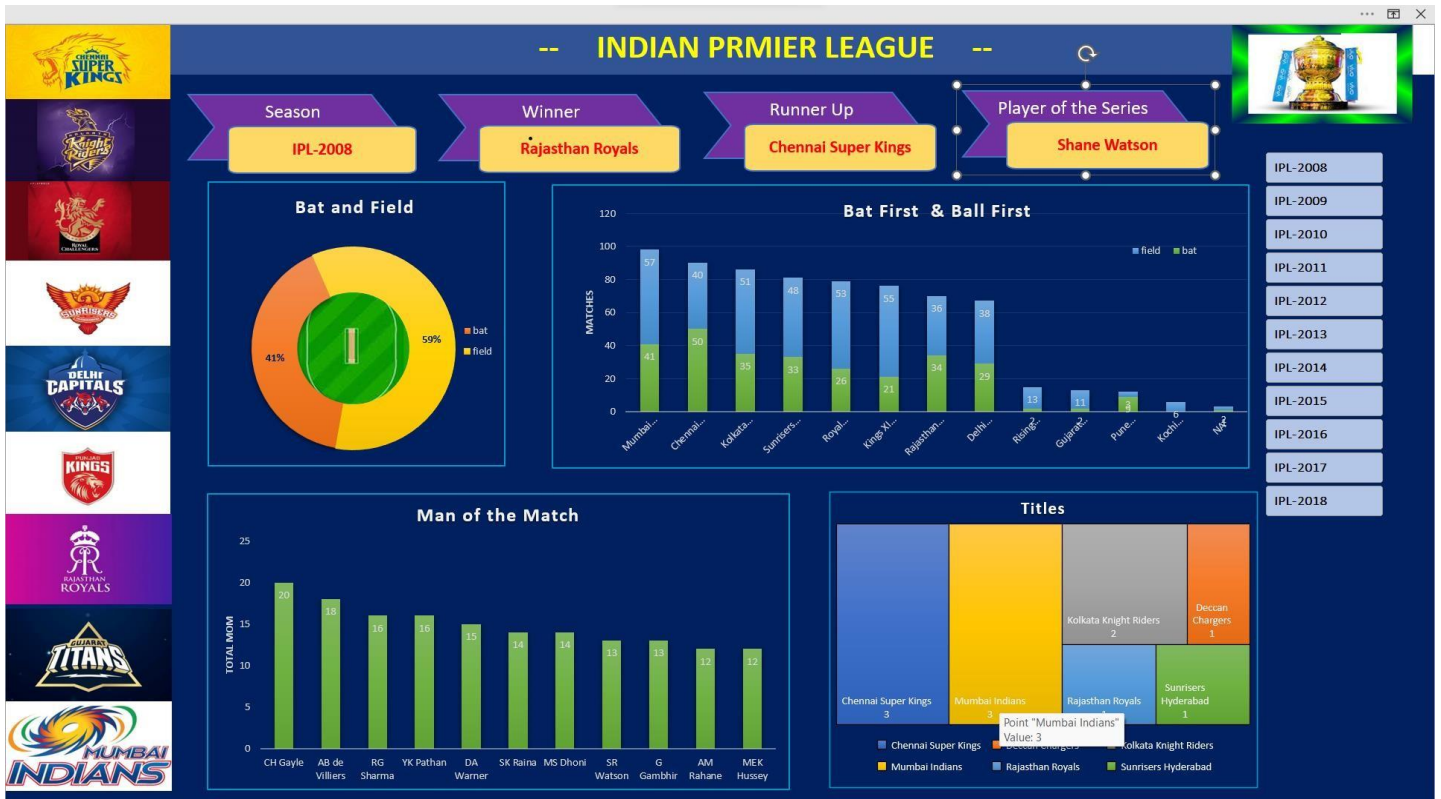
Tree and map chart.

Slicers are used to connect multiple pivot tables to make dynamic dashboard.

Venues Stats : It Given information that in present venue team wind against bat all field first



FINAL DASHBAORD



Future Scope

The Indian Premiere league is one of the best entertaining tournament of the world. Every Year of IPL come up with a new launching style.

IPL to me looks a very good option as far as providing the exposure to the young lot is concerned having played with the biggies of the games likes of Sachin, Kumble, Ponting etc. still hanging around with teams. As far as the skills is concerned, it does not dig deep into skill-testing especially for the batsmen, bowlers can still show their intelligence and skill by sticking to basics and containing the batter.

But with IPL the major issue is the crowd, since it takes place in major indian cities which are already hosts of test matches and some One-dayers it provides the crowd the option to choose between matches. So this decreases the attendance in the stadium during IPL since the crowd wants to see more or tough competitive cricket rather than friendly ventures.

And coming to the future, it has great scope and potential and with BCCI taking care of it so well it does not seem the trophy is gonna lose any shine in near future.

Reference and Bibliography

Youtube

Github

Wikipedia.com

Ipl.com

Google.com